

Assignment 4

Aaron Bussche

David Magaram

Jason Kasdiran

Group 28

April 14, 2026

Todo list

GLOBAL: target grade 8+. Rubric-driven checklist below. Remove todos before submit.	2
Introduction is too thin. State an explicit research question: “How do mutation rate and selection pressure shape the exploration–exploitation trade-off in a string-search GA, and does adding 2-opt local search (MA) outperform a plain EA on TSP under a <i>fair</i> (wall-clock) comparison?” Also: one sentence on why the exploration/exploitation trade-off matters, and one sentence previewing the main finding (so the reader knows where the report is going).	3
Add a “Reproducibility” paragraph at the end of Methods: language/version (Python X.Y), RNG seeding strategy, how many runs, where the code lives (repo URL or zip), and how to reproduce each figure (e.g. “run <code>python b1_run.py -config b1.yaml</code> to regenerate Fig. 1”). Assignment explicitly requires this.	3
Add one sentence explicitly stating the diversity measures used (Hamming mean pairwise distance on a subsample; position-wise Shannon entropy $H_i = -\sum_j f_{ij} \log f_{ij}$ averaged over positions), as required by the assignment appendix. Currently this is only visible via figure captions.	3
State the stopping criterion precisely: “A run is recorded as $t_{\text{finish}} = G_{\text{max}}$ (right-censored) when the target is not reached.” This matters for how the beeswarms should be read at $\mu = 0$ and $\mu = 3/L$	3
Caption of this table is fine. BUT: the report mentions $G_{\text{max}} = 200$ in Fig. 2’s caption while the parameter table says $G_{\text{max}} = 100$. Pick one and make them consistent, or explicitly note that the cap differs between experiments and why.	3
Justify the chosen parameters ($N = 100$, $K = 5$, $p_c = 0.8$, $\mu = 0.3$): one sentence citing the lecture defaults or a quick pilot run. The rubric explicitly asks whether methods are justified.	4
Describe the custom 50-city instance (how generated — uniform random in a unit square? seed?) and which TSPLIB instance was used (eil51). Currently unclear — a reader cannot reproduce without the code.	4
FIX CAPTION: should be “TSP EA/MA Parameters”, not “String Search GA Parameters” — copy-paste error. Also fix dataset typo “eli51” → “eil51”.	4
Add a row for the stopping criterion / generation budget and for the wall-clock budget used in Fig. 8/9. Also state the 2-opt termination rule (“until no improving 2-swap remains”) already mentioned in prose — good, but make it a table row too for quick reference.	4
Results currently consists of figures with captions doing most of the work. Rubric (section B “Description”) penalises this. For each figure, add 2–3 sentences of narrative in the main text that (a) references the figure as “Fig. ??”, (b) states what was measured, (c) states the observation, (d) interprets it. This is the single highest-leverage change for the grade.	5
Add a short paragraph introducing B1: what question this subsection answers (“Does the mutation rate μ trade off convergence speed against diversity, and does tournament size K change the optimum?”). Currently the reader lands straight on a figure.	5
Report quantitative results numerically as well, not only as curves: mean $\pm$ std of t_{finish} across the 10 runs for each condition, and number of runs that reached the target vs were censored at G_{max} . A small summary table will directly answer B.1 Q1–Q2 and Q6.	5

Caption is too terse. Spell out axes, the baseline ($K = 2$, $\mu = 1/L$, $N = 200$), what the bands/lines represent (mean best fitness with shaded $\pm$ std across 10 runs?), and the take-home (“all 10 runs find the target within $\sim X$ generations”). Same applies to every other figure below — captions should be self-contained.	5
B.1 Q2 explicitly asks for a beeswarm of t_{finish} distributions across the 10 runs per μ . If this figure is a fitness curve (as the caption suggests), ADD a second panel / figure with the beeswarm of t_{finish} (censored at G_{max} shown as e.g. hollow points). The assignment is specific about the plot type.	6
Fix typo “entrophy” $\rightarrow$ “entropy”. Specify what “diversity” is on the y-axis (mean pairwise Hamming distance / mean position-wise Shannon entropy in nats or bits?). Clarify whether this is averaged over the 10 runs and whether the band is std or min/max.	6
MISSING FIGURE for B.1 Q5: a plot of μ vs t_{finish} over ~ 10 μ values from 0 to $10/L$, with the optimum annotated. The parameter table already promises this sweep — deliver the figure and state the optimal μ in prose (e.g. “ $\mu^* \approx 1/L$ for $K = 2$, shifting toward $\approx 2/L$ for $K = 5$ ”).	7
Add a short “B1 take-away” paragraph after the figures: (1) optimal $\mu \approx 1/L$; (2) too low $\mu \rightarrow$ premature convergence (diversity collapse visible in Fig. 3); (3) too high $\mu \rightarrow$ random walk, never converges; (4) higher K shifts tolerable μ upward (more selection pressure compensates for more noise). This directly answers B.1 Q4 and Q6.	7
Add a short intro paragraph before the figures: restate the research question (“does 2-opt local search improve TSP performance beyond the extra compute it costs?”), name the two datasets, and preview that the fair-comparison answer requires wall-clock normalisation (you already argue this below — move the argument up so the figures land with context).	7
Caption should name the datasets on each panel, the y-axis (mean best tour length in same units as the coordinates?), and what the shaded band is. Also state the gap numerically (e.g. MA $\approx X$, EA $\approx Y$ on eil51; known optimum for eil51 is 426, mention it).	7
“Consistency” — quantify. Report std or IQR of final tour length across 10 runs for each (dataset, algorithm). Consider a Mann–Whitney U test on final tour lengths to support the claim with a p -value; the rubric (“Validity”) rewards statistical backing of claims.	8
This is the core conclusion of B.2 — strengthen it with numbers: “On eil51, MA reached mean tour length $X \pm s_X$ vs EA’s $Y \pm s_Y$ within the same wall-clock budget of T s (Mann–Whitney U, $p < 0.01$).” Without numbers this reads as an assertion.	9
Note the two paragraphs above (the “We considered. . .” block and the “Comparing the MA. . .” block) are near-duplicates of each other — one is clearly an earlier draft. Keep the better one (the second is more polished) and delete the other.	9
EMPTY. The rubric (“Clarity” + assignment “take-home message”) makes this section worth real points. Write 4–6 sentences covering: (1) the main finding on B1 — optimal $\mu \approx 1/L$ balances exploration and exploitation, and higher K shifts this upward; (2) the main finding on B2 — MA beats EA even under a fair wall-clock comparison because 2-opt escapes local optima; (3) one-sentence take-home: “local search is worth its compute cost for TSP; for string search there is a narrow optimal mutation rate that depends on selection pressure.”	9
EMPTY. Add 4–6 sentences. Must cover: (1) limitations — wall-clock is hardware/implementation-dependent, single machine, 10 runs is modest, only two TSP instances tested, only one target string; (2) threats to validity — 2-opt is specialized to TSP so the MA advantage may not generalize; censoring of t_{finish} at G_{max} biases the mean downward; (3) future work — test on larger TSPLIB instances, vary 2-opt intensity (full vs. k steps, Baldwinian vs Lamarckian), try adaptive μ . The rubric explicitly mentions “discuss some potential limitations”.	10
MISSING FIGURES. Assignment A.2 requires: (i) a plot of best fitness vs. generations over 1500 generations for the (1+1)-GA with selection (Q1); (ii) the same plot with the no-selection variant overlaid (Q2); (iii) a figure demonstrating the performance difference using multiple runs — e.g. mean $\pm$ std across say 20 runs, or box plots of final fitness (Q4). Text answers below are fine but unsupported without these plots.	11

GLOBAL: target grade 8+. Rubric-driven checklist below. Remove todos before submit.

Introduction

Introduction is too thin. State an explicit research question: “How do mutation rate and selection pressure shape the exploration–exploitation trade-off in a string-search GA, and does adding 2-opt local search (MA) outperform a plain EA on TSP under a *fair* (wall-clock) comparison?” Also: one sentence on why the exploration/exploitation trade-off matters, and one sentence previewing the main finding (so the reader knows where the report is going).

Evolutionary algorithms (EA) are a population-based optimization method. They are based on natural selection and work by exploring new solution areas and refining good already found solutions, where candidate solutions are improved by mutating and crossover.

By investigating the traveling salesman problem and string search, this project explores how different parameter configurations influence convergence speed, population diversity, and algorithm performance.

Methods

Add a “Reproducibility” paragraph at the end of Methods: language/version (Python X.Y), RNG seeding strategy, how many runs, where the code lives (repo URL or zip), and how to reproduce each figure (e.g. “run `python b1_run.py -config b1.yaml` to regenerate Fig. 1”). Assignment explicitly requires this.

This project will implement a genetic algorithm (GA) for string search and memetic algorithm (MA) for traveling salesman problem. This section describes the experimental setup and design choices.

String search

A string search genetic algorithm was implemented and the used parameters are displayed in Table 1. “NaturalComputing” was used as the target string with a length $L = 16$, comprising of both lowercase and uppercase letters from an alphabet $|\Sigma|$. Since fitness is the number of characters that match the target, the maximum fitness is 16.

The algorithm aims to match the target string by evolving a population of $N = 200$ individuals. Each generation follows a few simple steps. First, during the tournament selection phase, a random subset K is drawn from the population N and the fittest string is kept for reproduction.

Larger K values mean more potentially high-fitness strings, which can lead to the selection of only the best performers, working as a filter against weaker candidates. While it usually leads to faster convergence, it can also reduce diversity as the entire population will mimic the strongest strings.

Then, each pair undergoes single-point crossover with a probability $p_c = 1$, producing 2 new offsprings. Crossover selects a random index and swaps the parent’s characters beyond that cutoff. Additionally, every character of the offspring has a small mutation rate μ to randomly become one of the letters from the alphabet $|\Sigma|$. Finally, the entire parent generation is replaced with the new offspring generation.

All algorithms were run single-threaded on a desktop system with a AMD Ryzen 5800x processor.

Add one sentence explicitly stating the diversity measures used (Hamming mean pairwise distance on a subsample; position-wise Shannon entropy $H_i = -\sum_j f_{ij} \log f_{ij}$ averaged over positions), as required by the assignment appendix. Currently this is only visible via figure captions.

State the stopping criterion precisely: “A run is recorded as $t_{\text{finish}} = G_{\text{max}}$ (right-censored) when the target is not reached.” This matters for how the beeswarms should be read at $\mu = 0$ and $\mu = 3/L$.

Caption of this table is fine. BUT: the report mentions $G_{\text{max}} = 200$ in Fig. 2’s caption while the parameter table says $G_{\text{max}} = 100$. Pick one and make them consistent, or explicitly note that the cap differs between experiments and why.

Table 1: String Search GA Parameters

Parameter	Value
String length L	16
Alphabet $ \Sigma $	52
Population size N	200
Tournament selection K	$\{2, 5\}$
Crossover probability p_c	1
Mutation rate μ	$\{0, 1/L, 3/L\} + \text{sweep from } 0/L \text{ to } 10/L$
Runs	10
Max generations $G_{\max}$	100

Table 2: Parameters for Exercise B1.

Traveling Salesman Problem

Justify the chosen parameters ($N = 100$, $K = 5$, $p_c = 0.8$, $\mu = 0.3$): one sentence citing the lecture defaults or a quick pilot run. The rubric explicitly asks whether methods are justified.

Describe the custom 50-city instance (how generated — uniform random in a unit square? seed?) and which TSPLIB instance was used (*eil51*). Currently unclear — a reader cannot reproduce without the code.

For the TSP, both a simple evolutionary algorithm (EA) and a memetic algorithm (MA) were implemented. Each solution is encoded as a permutation of city indices representing a tour. The simple EA uses ordered crossover (OX): a random contiguous segment is copied from one parent and the remaining cities are filled in the order they appear in the second parent. Mutation is applied via swap mutation, where two randomly selected positions in the tour are exchanged. Tournament selection ($K = 5$) is used to select parents, and generational replacement with no elitism is applied.

The memetic algorithm extends the EA by applying 2-opt local search to each individual after mutation. The 2-opt procedure iteratively reverses sub-sequences of the tour to eliminate crossing edges, repeating until no improving swap can be found. This Lamarckian approach allows individuals to locally optimize their fitness before evaluation, which typically accelerates convergence. Both algorithms were evaluated on the custom 50-city dataset and on the TSPLIB benchmark instance *eil51*.

Table 3: String Search GA Parameters

FIX CAPTION: should be “TSP EA/MA Parameters”, not “String Search GA Parameters” — copy-paste error. Also fix dataset typo “eli51” → “eil51”.

Parameter	Value
Population size N	100
Tournament selection K	5
Crossover probability p_c	0.8
Mutation rate μ	0.3
Local search	2-opt
Runs	10
Datasets	eil51, TSPLIB, Custom 50-city instance

Table 4: Parameters for Exercise B2.

Add a row for the stopping criterion / generation budget and for the wall-clock budget used in Fig. 8/9. Also state the 2-opt termination rule (“until no improving 2-swap remains”) already mentioned in prose — good, but make it a table row too for quick reference.

Results

Results currently consists of figures with captions doing most of the work. Rubric (section B “Description”) penalises this. For each figure, add 2–3 sentences of narrative in the main text that (a) references the figure as “Fig. ??”, (b) states what was measured, (c) states the observation, (d) interprets it. This is the single highest-leverage change for the grade.

B1

Add a short paragraph introducing B1: what question this subsection answers (“Does the mutation rate μ trade off convergence speed against diversity, and does tournament size K change the optimum?”). Currently the reader lands straight on a figure.

Report quantitative results numerically as well, not only as curves: mean $\pm$ std of t_{finish} across the 10 runs for each condition, and number of runs that reached the target vs were censored at G_{max} . A small summary table will directly answer B.1 Q1–Q2 and Q6.

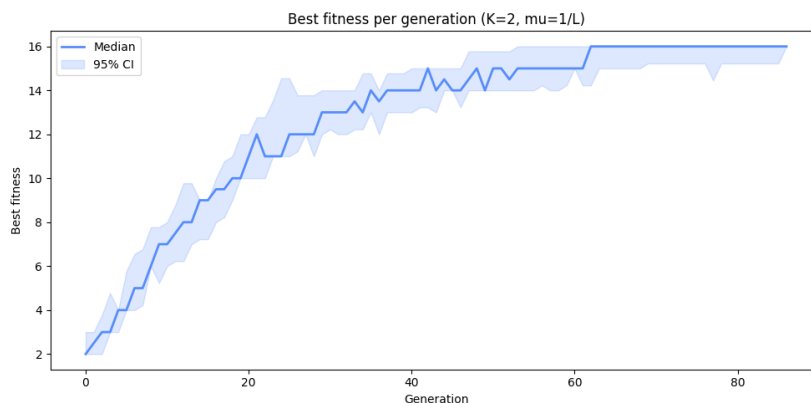


Figure 1: Results of running the GA 10 times with baseline values

Caption is too terse. Spell out axes, the baseline ($K = 2$, $\mu = 1/L$, $N = 200$), what the bands/lines represent (mean best fitness with shaded $\pm$ std across 10 runs?), and the take-home (“all 10 runs find the target within $\sim X$ generations”). Same applies to every other figure below — captions should be self-contained.

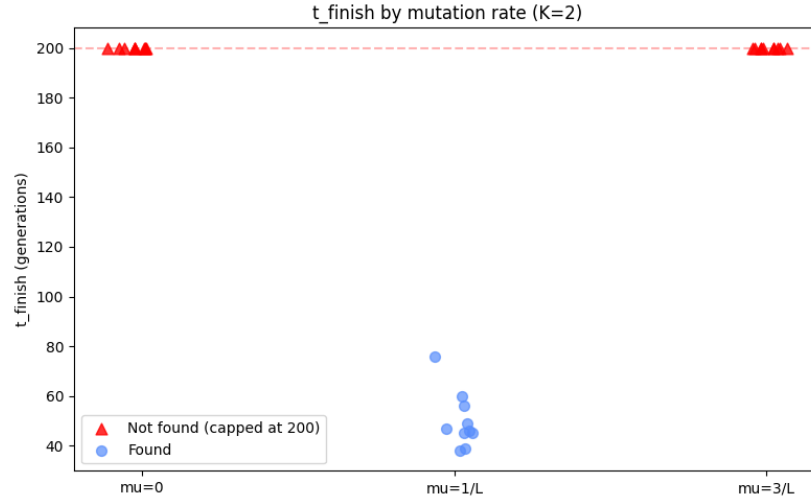


Figure 2: Three different mutation rates with $K=2$. Of these mutation rates, the GA manages to find the target only with $\mu=1/L$ before getting cut off at 200 Generations

B.1 Q2 explicitly asks for a beeswarm of t_{finish} distributions across the 10 runs per μ . If this figure is a fitness curve (as the caption suggests), ADD a second panel / figure with the beeswarm of t_{finish} (censored at G_{max} shown as e.g. hollow points). The assignment is specific about the plot type.

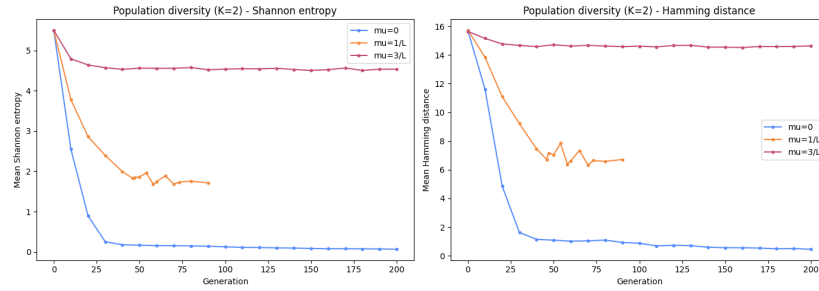


Figure 3: Measuring the diversity of the runs in two different ways, with Shannon entropy and Hamming distance. Both show that $\mu=0$ leads to no diversity, $\mu=3/L$ leads to high diversity through randomness while $\mu=1/L$ manages to strike a middle ground (and reach the target).

Fix typo “entropy” $\rightarrow$ “entropy”. Specify what “diversity” is on the y-axis (mean pairwise Hamming distance / mean position-wise Shannon entropy in nats or bits?). Clarify whether this is averaged over the 10 runs and whether the band is std or min/max.

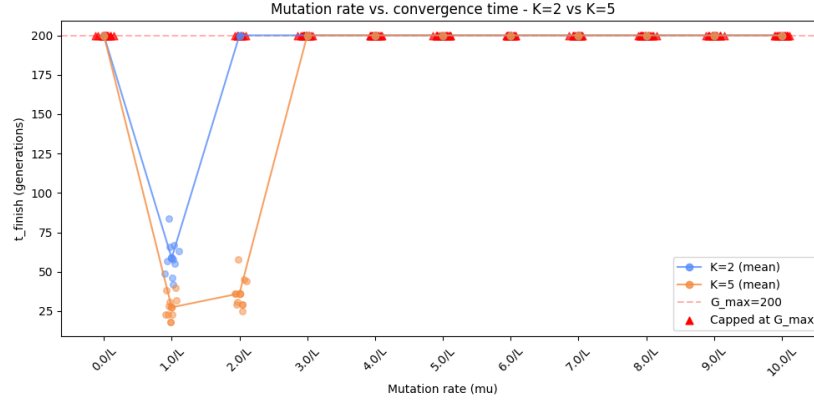


Figure 4: Higher tournament size increases selection pressure. We repeat the μ sweep to see how this interacts with mutation rate. $K=5$ leads to results faster & can also converge with $\mu=2/L$, though $\mu=1/L$ remains optimal.

MISSING FIGURE for B.1 Q5: a plot of μ vs t_{finish} over ~ 10 μ values from 0 to $10/L$, with the optimum annotated. The parameter table already promises this sweep — deliver the figure and state the optimal μ in prose (e.g. “ $\mu^* \approx 1/L$ for $K = 2$, shifting toward $\approx 2/L$ for $K = 5$ ”).

Add a short “B1 take-away” paragraph after the figures: (1) optimal $\mu \approx 1/L$; (2) too low $\mu \rightarrow$ premature convergence (diversity collapse visible in Fig. 3); (3) too high $\mu \rightarrow$ random walk, never converges; (4) higher K shifts tolerable μ upward (more selection pressure compensates for more noise). This directly answers B.1 Q4 and Q6.

B2

Add a short intro paragraph before the figures: restate the research question (“does 2-opt local search improve TSP performance beyond the extra compute it costs?”), name the two datasets, and preview that the fair-comparison answer requires wall-clock normalisation (you already argue this below — move the argument up so the figures land with context).

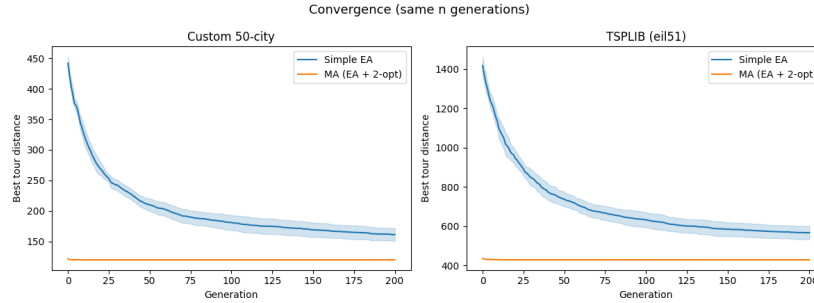


Figure 5: Both of the datasets show MA reaching better results in fewer generations, and in total.

Caption should name the datasets on each panel, the y-axis (mean best tour length in same units as the coordinates?), and what the shaded band is. Also state the gap numerically (e.g. $MA \approx X$, $EA \approx Y$ on eil51; known optimum for eil51 is 426, mention it).

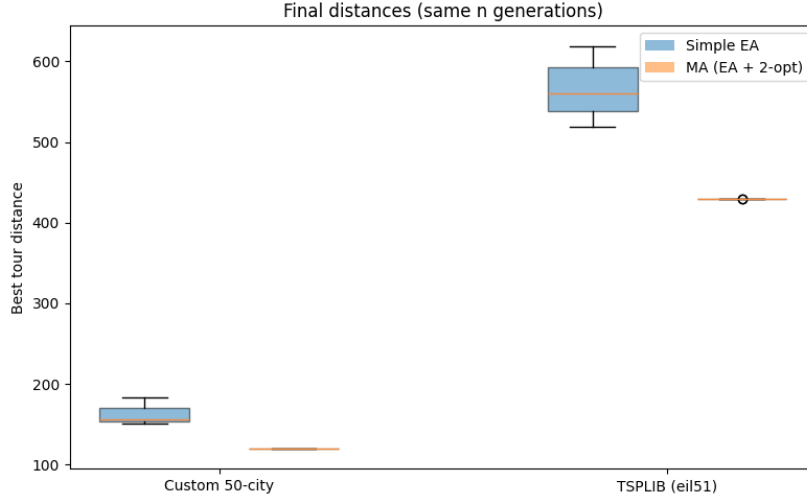


Figure 6: In this comparison, MA clearly beats EA in distance and consistency.

“Consistency” — quantify. Report std or IQR of final tour length across 10 runs for each (dataset, algorithm). Consider a Mann–Whitney U test on final tour lengths to support the claim with a p -value; the rubric (“Validity”) rewards statistical backing of claims.

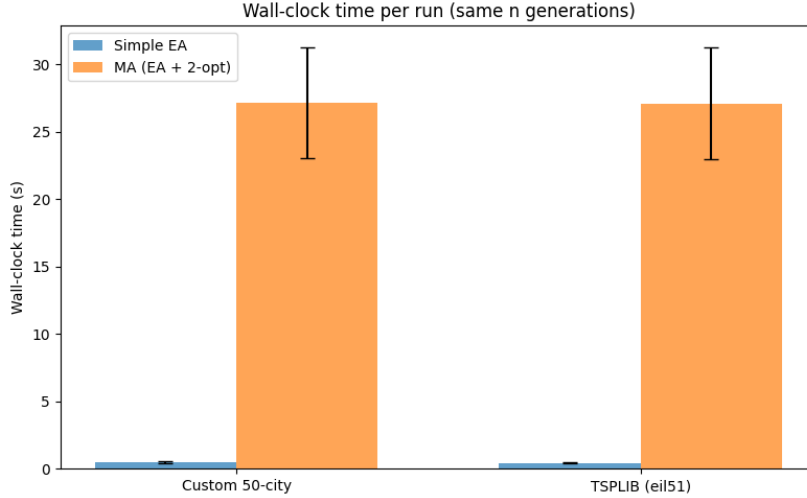


Figure 7: Per generation, MA takes much more compute & longer than EA. It is therefore not fair to compare them in this way.

We considered comparing them on number of operations, but the operations of MA are only in its neighborhood, so comparing them directly with whole generations of EA would again be unfair. We therefore decided to compare wall-clock time that, while influenceable by all kinds of factors and therefore not overly scientific, provides a good baseline for real-use performance, especially if averaged over 10 runs each.

Comparing the MA and simple EA on the same number of generations is not a fair baseline. Each MA generation includes a full 2-opt local search over every individual in the population, making it far more computationally expensive than a single EA generation. Any advantage observed per generation could therefore reflect additional computation rather than algorithmic superiority. We considered normalizing by number of fitness evaluations, but since 2-opt operates within a solution’s neighborhood its evaluation count is not directly comparable to a full population generation in the EA. We therefore compare on wall-clock time, which — while sensitive to system-level factors — provides a practical performance baseline when averaged over 10 runs.

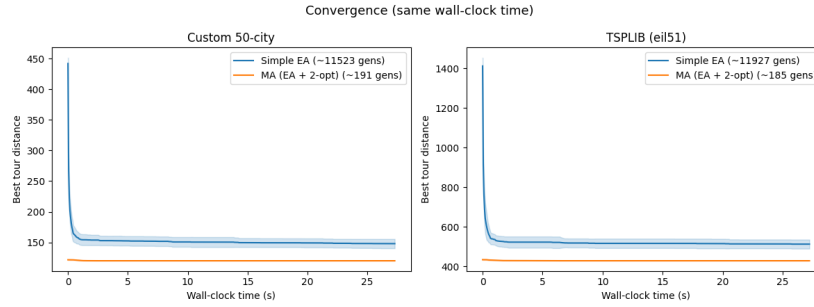


Figure 8: If compared instead on wall-clock time, MA still converges on better values, but EA’s curve also flattens to near its best result in a comparable time as MA. While EA goes through more than 11-thousand generations in this time, MA only reaches around 190.

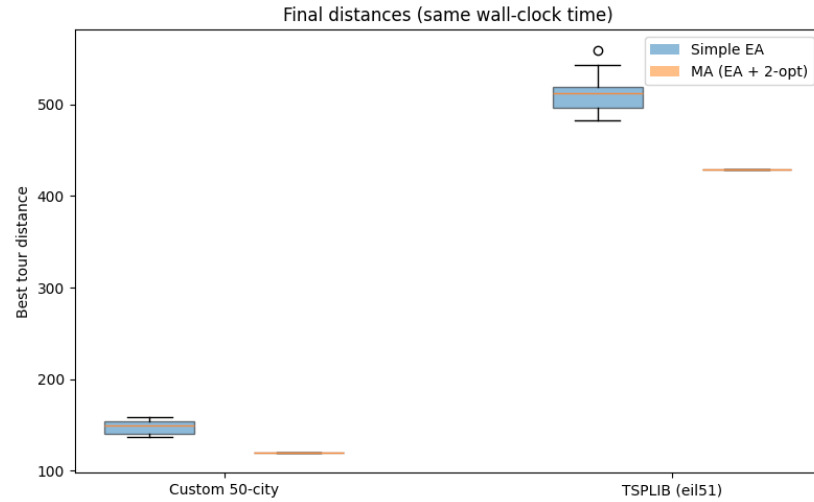


Figure 9: While the gap has shrunk considerably from comparing by number of generations, MA still provides better and more consistent results than EA.

Even under a fair wall-clock comparison, the MA consistently reaches shorter and more consistent tour distances than the simple EA on both datasets. The addition of local search therefore genuinely improves performance beyond what the extra computation alone can explain: 2-opt allows individuals to escape local optima that trap the plain EA, reducing both mean tour length and variance across runs.

This is the core conclusion of B.2 — strengthen it with numbers: “On eil51, MA reached mean tour length $X \pm s_X$ vs EA’s $Y \pm s_Y$ within the same wall-clock budget of T s (Mann–Whitney U , $p < 0.01$).” Without numbers this reads as an assertion.

Note the two paragraphs above (the “We considered...” block and the “Comparing the MA...” block) are near-duplicates of each other — one is clearly an earlier draft. Keep the better one (the second is more polished) and delete the other.

Conclusion

EMPTY. The rubric (“Clarity” + assignment “take-home message”) makes this section worth real points. Write 4–6 sentences covering: (1) the main finding on B1 — optimal $\mu \approx 1/L$ balances exploration and exploitation, and higher K shifts this upward; (2) the main finding on B2 — MA beats EA even under a fair wall-clock comparison because 2-opt escapes local optima; (3) one-sentence take-home: “local search is worth its compute cost for TSP; for string search there is a narrow optimal mutation rate that depends on selection pressure.”

Discussion

EMPTY. Add 4–6 sentences. Must cover: (1) limitations — wall-clock is hardware/implementation-dependent, single machine, 10 runs is modest, only two TSP instances tested, only one target string; (2) threats to validity — 2-opt is specialized to TSP so the MA advantage may not generalize; censoring of t_{finish} at G_{max} biases the mean downward; (3) future work — test on larger TSPLIB instances, vary 2-opt intensity (full vs. k steps, Baldwinian vs Lamarckian), try adaptive μ . The rubric explicitly mentions “discuss some potential limitations”.

Appendix

Part A

A1

What can you conclude about the effects of fitness scaling on selection pressure?

Adding constants raises all fitness all values by that amount. So it smooths out all the values. For example, when we compare f2 (which only squares X) and f4 (which squares X and adds 20), we can see that the probabilities of f4 are closer together. Probabilities are increased and reduced, so there is a much smaller gap between them, which also reduces selection pressure. Multiplying by a constant does not change the selection probabilities. If we compare f2 and f3, we can see that the probabilities stay the same even if we scale it. So scaling fitness does not have an effect on the selection pressure.

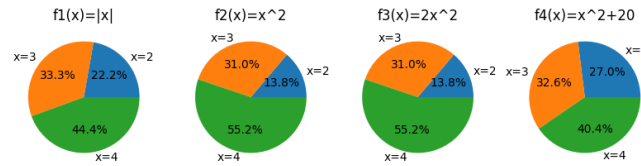


Figure 10: Selection probability of the three individuals for each function.

x	f1	p1	f2	p2	f3	p3	f4	p4
2	2	0.2222	4	0.1379	8	0.1379	24	0.2697
3	3	0.3333	9	0.3103	18	0.3103	29	0.3258
4	4	0.4444	16	0.5517	32	0.5517	36	0.4045

Table 5: Fitness function value and selection probability for each individual and each function.

A2

MISSING FIGURES. Assignment A.2 requires: (i) a plot of best fitness vs. generations over 1500 generations for the (1+1)-GA with selection (Q1); (ii) the same plot with the no-selection variant overlaid (Q2); (iii) a figure demonstrating the performance difference using multiple runs — e.g. mean $\pm$ std across say 20 runs, or box plots of final fitness (Q4). Text answers below are fine but unsupported without these plots.

Does the algorithm find the optimum? Yes. It is able to reach fitness 100, meaning all 100 bits were successfully set to 1.

What do you see? The version without selection is stuck at around 50 and never converges. The version with selection keeps steadily climbing, ultimately reaching 100.

Can you conclude anything on the difference between these two versions based on the above results? If yes, why? If not, what should you do to make a fair comparison? No. We would need multiple runs as EAs are stochastic and each run produces different results because of initialization and random mutation. To make it fair, we should run each version multiple times and use averages.

Is there a difference in performance? Yes. It is apparent that there is a difference in performance between the versions. Across all the runs, the version with selection always converges to fitness 100 while the version without selection is always near 50 through all the 1500 generations. The algorithm has no memory of good solutions and cannot improve.

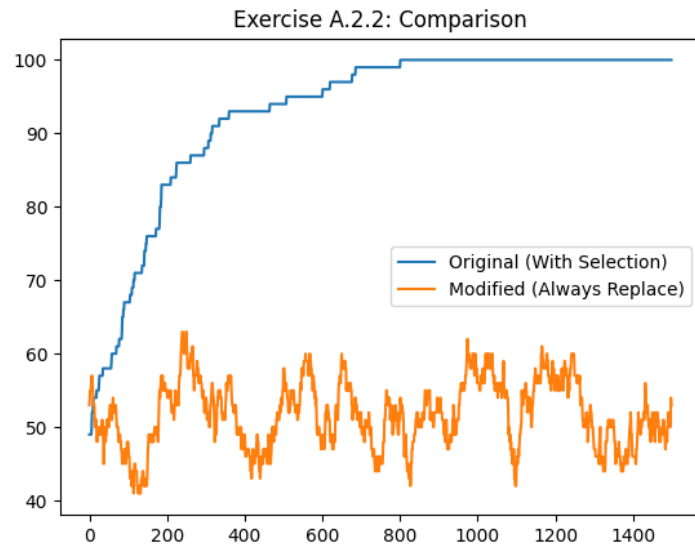


Figure 11: Fitness against generations.

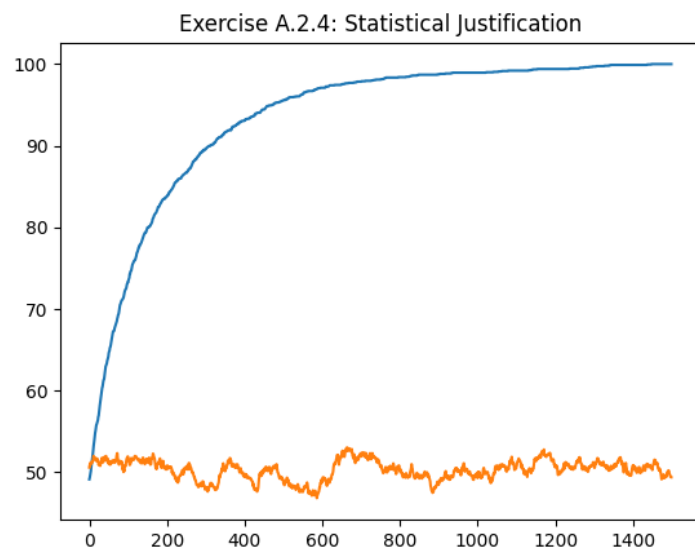


Figure 12: Averages of fitness vs generations across multiple runs.